Pranav V Bhat

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Education

PES University, Bachelor of Technology (B.Tech), Computer Science — 2023 – Expected Jul 2027

- **Coursework:** Computer Networks, Operating Systems, DSA, Cloud Computing, Web Technologies, DBMS, Machine Learning
- CGPA 8.3/10.0 (through Semester 6); **CNR Rao Scholarship** (top 20% of cohort); **Google Cybersecurity Certificate**

Experience

Karplexus, SWE Intern — Jun 2025 – Aug 2025

- Built a containerized log-triage service (Python, FastAPI REST APIs, OpenSearch, Docker) to detect and classify critical events.
- Integrated LLMs into the pipeline to classify log events and generate suggested responses, automating a manual triage step.
- Built a Slack bot and FastAPI alert-routing path that delivers targeted triage steps to SRE teams, to cut manual investigation.

GoWarm, Developer — Mar 2025 – May 2025

- Tested a web scraper against anti-bot security controls in local and AWS environments, documenting edge cases and failure modes.
- Added rotating-VPN support to an AWS-hosted scraper and validated its resilience as access controls changed.

ACM PESUECC Chapter, Member and Open-Source Maintainer — Aug 2024 – Present

- On the organizing team for Hacknight, a Hacktoberfest hackathon; maintain open-source repositories for students.
- Delivered a workshop on Git fundamentals (version-control workflows, repository management); published the guide online.

PES University, Python Teaching Assistant — Sep 2026 – Present

- Explain Python concepts and debugging strategy to students, guiding them to break problems into steps.

Projects

Ballrs: Layer 7 Reverse Proxy and Load Balancer in Rust — Jul 2024

- Designed a Layer 7 reverse proxy and load balancer (Tokio, Hyper) with six traffic-routing algorithms (incl. weighted), YAML config profiles, connection limits and request timeouts; design write-up.
- Implemented concurrent per-server **health checks** and automatic **failover** that reroute requests away from failed servers.
- Benchmarked all six algorithms (126 measured runs, zero failed requests) against a direct-backend reference on loopback: GET reached **40.0–41.6k req/s** at concurrency 32 (93.5–97.4% of direct); flagged a 64 KiB streamed-POST tail-latency anomaly (p95 1–42 ms) for profiling.

radFS: In-Memory FUSE Filesystem in Go (Co-Lead, Mentor) — 2026

- Co-led and mentored four mentees building a userspace FUSE/Go filesystem that prioritizes memory density over speed.
- Chose Adaptive Radix Trees over B-Trees to address worst-case memory use, trading raw speed for density.
- As code owner, reviewed mentee pull requests, including AI-generated code; on a 57-commit ART-integration PR, flagged deadlock and key-aliasing risks, added regression tests (ownership, timestamps, duplicate-create errors) and merged.

MinionFS: Union Filesystem in Go (FUSE) — 2026

- Built an overlay filesystem (copy-on-write, whiteout deletes) with pluggable AES-256-GCM encryption and gzip codecs.
- Wrote Go unit tests (directories, files, codec round-trips) with coverage reporting and a shell test script; Justfile automation.

Rustbin: Concurrent File-Sharing API in Rust — Nov 2024

- Built a file-sharing HTTP API (Tokio, Warp) with asynchronous multipart streaming and non-blocking file I/O.
- Added semaphore-based concurrency limits, streaming data paths and payload caps for predictable resource use under load.

Skills

Languages: Rust, Go, Python, C/C++, JavaScript, Java, SQL, Bash

Networking and Back-end: TCP, HTTP/1.1, Layer 7 load balancing, failover, REST APIs, gRPC, Tokio, Hyper, Warp, FastAPI, async I/O, AES-GCM

Cloud, Data and DevOps: AWS, Docker, Kubernetes, GitHub Actions (CI/CD), Linux/Unix, OpenSearch, SQL/DBMS, Git, FUSE

AI and Quality: GitHub Copilot, OpenAI Codex, reviewing AI-generated code, LLM log classification, unit and load testing